

Dynamic Stochastic General Equilibrium and Business Cycles

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Research Track Master

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CHAPTER 3

Perturbation Methods and the Full-fledged RBC Model

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HANDOUT OBJECTIVES

- Understanding the Real Business Cycles and its theoretical foundations regarding consumption smoothing, asset price fluctuations etc.
- Developing different models of asset pricing.
- Performing higher order approximations to the model's policy function (second or third order IRF).
- Estimating the RBC model with four time series.

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1. A full-fledged RBC model

This model is inspired by the business cycle model of Hansen (1985) which aims at fixing anomalies in the real business cycle theory of Lucas (1977) and Kydland and Prescott (1982). Before Hansen, the intratemporal substitution between consumption and leisure was unable to explain large aggregate fluctuations in hours worked/employment.

1.1 The household

Our economy is populated by identical and infinite living households distributed over a continuum $j \in [0, 1]$. The representative household utility function reads as follows:

$$\mathcal{U}(C_{jt}, H_{jt}) = \frac{C_{jt}^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{jt}^{1+\sigma_L}}{1+\sigma_L}$$

Utility is increasing in consumption C_{jt} and decreasing in labor H_{jt} where σ^C and σ^L are curvature parameters. Households in this economy maximise the expected value of their utility:

$$\max_{\{C_{jt}, H_{jt}, B_{jt}, K_{jt+1}, I_{jt}\}} \mathbb{E}_t \sum_{\tau=0}^{\infty} \beta^\tau \varepsilon_{t+\tau}^C \mathcal{U}(C_{jt}, H_{jt}), \quad (1)$$

where β is the discount factor affected by an exogenous AR(1) preference shock ε_t^C . Households face an intertemporal problem, they find the value of consumption C_{jt} , hours worked H_{jt} and bonds B_{jt} which delivers the highest value of utility. They face the following budget constraint which binds every period:

$$W_t H_{jt} + Z_t K_{jt-1} + R_{t-1} B_{jt-1} = C_{jt} + I_{jt} + B_{jt} + T_{jt}, \quad (2)$$

where W_t is the real wage, Z_t the remuneration of capital services, R_{t-1} is the interest rate for bonds subscribed in $t-1$ and T_{jt} are slump sum taxes which aim at financing government budget. The law of motion of capital also bounds the household's optimization problem:

$$K_{jt} = \varepsilon_t^I I_{jt} + (1 - \delta) K_{jt-1}, \quad (3)$$

where K_{jt} is the stock of capital bought in t (the lag is explained by the time-to-build hypothesis regarding capital setting, *i.e.* physical capital requires one period to be installed), I_{jt} denotes investment in t while parameter δ denotes the depreciation rate of physical capital. The lagrangian problem is given by:

$$\mathcal{L}_t = \mathbb{E}_t \sum_{\tau=0}^{\infty} \varepsilon_{t+\tau}^C \beta^\tau \left[\frac{C_{jt+\tau}^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{jt+\tau}^{1+\sigma_L}}{1+\sigma_L} + \lambda_{t+\tau}^c \left[W_{t+\tau} H_{jt+\tau} + Z_{t+\tau} K_{jt-1+\tau} + R_{t-1+\tau} B_{jt-1+\tau} - C_{jt+\tau} - I_{jt+\tau} - B_{jt+\tau} - T_{jt+\tau} \right] + \lambda_{t+\tau}^k \left[\varepsilon_{t+\tau}^I I_{jt+\tau} + (1 - \delta) K_{jt-1+\tau} - K_{jt+\tau} \right] \right]$$

where λ_t^c et λ_t^k are lagrange multipliers associated to the budget constraint and the physical capital accumulation constraint. The first order conditions are given by the welfare maximisation in C_{jt} , H_{jt} , I_{jt} , K_{jt} and B_{jt} :

$$\begin{aligned} (\partial C_{jt}) : \lambda_t^c &= C_{jt}^{-\sigma_C} \\ (\partial H_{jt}) : \chi H_{jt}^{\sigma_L} &= \lambda_t^c W_t \\ (\partial B_{jt}) : \mathbb{E}_t \left\{ \frac{e^{\varepsilon_t^C} \lambda_t^c}{e^{\varepsilon_{t+1}^C} \lambda_{t+1}^c} \right\} &= \beta R_t \\ (\partial I_{jt}) : \lambda_t^c &= \varepsilon_t^I \lambda_t^k \\ (\partial K_{jt}) : \varepsilon_t^C \lambda_t^k &= \beta \mathbb{E}_t \left\{ \varepsilon_{t+1}^C \left[(1 - \delta) \lambda_{t+1}^k + \lambda_{t+1}^c Z_{t+1} \right] \right\} \end{aligned}$$

Substituting the lagrange multipliers, our optimal conditions boils down to an Euler equation:

$$\beta R_t = \mathbb{E}_t \left\{ \frac{\varepsilon_t^C}{\varepsilon_{t+1}^C} \left(\frac{C_{jt+1}}{C_{jt}} \right)^{\sigma_C} \right\}, \quad (4)$$

an hour supply equation:

$$W_t = \chi H_{jt}^{\sigma_L} C_{jt}^{\sigma_C} \quad (5)$$

and a no-arbitrage condition between bonds and assets:

$$\frac{R_t}{\varepsilon_t^I} = \mathbb{E}_t \left\{ \frac{(1 - \delta)}{\varepsilon_{t+1}^I} + Z_{t+1} \right\} \quad (6)$$

Regarding the labor supply equation, there are two effects that drive the labor in the utility-optimization framework. Using the household FOC on labor, the labor supply is given by $H_{jt} = [\lambda_t^c W_t / \chi]^{1/\sigma_L}$.

- XX effect: $\partial H_{jt} / \partial W_t = [\lambda_t^c W_t / \chi]^{1/\sigma_L} / (\sigma_L W_t) > 0$, so basically a rise in wage generate a rise in hours worked.

- wealth effect $\partial H_{jt} / \partial \lambda_t^c = [\lambda_t^c W_t / \chi]^{1/\sigma_L} / (\sigma_L \lambda_t^c) > 0$, similarly a rise in the wealth of households (measured by the marginal utility of consumption) rise the hours supply.

1.2 Firms

Firms are homogeneous and distributed on an unitary interval $i \in [0, 1]$ and have the following production technology:

$$Y_{it} = \varepsilon_t^A K_{it-1}^\alpha H_{it}^{1-\alpha} \quad (7)$$

where Y_{it} is the production, K_{it-1} and H_{it} are input factors, i.e. the quantity of physical capital and hours involved in the production process of firms. Finally, ε_t^A denotes the productivity shock process.

The optimisation problem can be expressed in two ways.

The first one can be expressed as a profit maximisation problem:

$$\max_{Y_{it}, K_{it-1}, H_{it}} Y_{it} - Z_t K_{it-1} - W_t H_{it} + \lambda_t [\varepsilon_t^A K_{it-1}^\alpha H_{it}^{1-\alpha} - Y_{it}]$$

where λ_t denotes the lagrange multiplier on the supply constraint, it can also interpreted as the marginal profit.

The second way can be viewed as an inputs minimisation problem:

$$\min_{K_{it}, H_{it}} Z_t K_{it-1} + W_t H_{it} - \lambda_t [\varepsilon_t^A K_{it-1}^\alpha H_{it}^{1-\alpha} - Y_{it}]$$

where λ_t denotes the lagrange multiplier on the supply constraint and can also be interpreted as the marginal cost of producing an additional unit of good.

First order conditions are:

$$\begin{aligned} (\partial K_{it-1}) : Z_t &= \alpha \lambda_t \frac{Y_{it}}{K_{it-1}} \\ (\partial H_{it}) : W_t &= (1 - \alpha) \lambda_t \frac{Y_{it}}{H_{it}} \\ (\partial Y_{it}) : \lambda_t &= 1 \end{aligned}$$

Replacing the lagrange multiplier, the two conditions which determine the highest profits/lowest costs path are:

$$Z_t = \alpha \frac{Y_{it}}{K_{it-1}} \quad (8)$$

$$W_t = (1 - \alpha) \frac{Y_{it}}{H_{it}} \quad (9)$$

1.3 Aggregation and Equilibrium conditions

Our economy is populated by two agents: households j and firms i . Their interactions, summarized by market clearing conditions for goods, labour and physical capital markets, can be expressed as follows:

$$\underbrace{\int_0^1 Y_{it} di}_{\text{aggregate supply}} = \underbrace{\int_0^1 C_{jt} dj}_{\text{aggregate consumption}} + \underbrace{\int_0^1 I_{jt} dj}_{\text{aggregate investment}} + \underbrace{\bar{G} \varepsilon_t^G}_{\text{aggregate public spending}}$$

We assume output is driven exogenously by demand shocks, such that $\bar{G} = g^y \bar{Y}$, where g^y is the fixed spending-to-GDP ratio. Variable ε_t^G denotes the demand shock which affects output.

Letting Y_t , C_t and I_t denote the aggregate supply, consumption and investment, the resource constraint boils down to:

$$Y_t = C_t + I_t + g^y \bar{Y} \varepsilon_t^G. \quad (10)$$

The equilibrium on the labour market is:

$$H_t = \underbrace{\int_0^1 H_{it} di}_{\text{labour demand}} = \underbrace{\int_0^1 H_{jt} dj}_{\text{labour supply}},$$

and for physical capital:

$$K_t = \underbrace{\int_0^1 K_{it} di}_{\text{capital demand}} = \underbrace{\int_0^1 K_{jt} dj}_{\text{capital supply}}.$$

1.4 Disturbances

Our shock processes are characterized by four disturbances (in logs):

$$\begin{aligned} \log(\varepsilon_t^A) &= \rho_A \log(\varepsilon_{t-1}^A) + \eta_t^A \quad \text{with } \eta_t^A \sim \mathcal{N}(0, \sigma_A^2) \\ \log(\varepsilon_t^G) &= \rho_G \log(\varepsilon_{t-1}^G) + \eta_t^G \quad \text{with } \eta_t^G \sim \mathcal{N}(0, \sigma_G^2) \\ \log(\varepsilon_t^C) &= \rho_C \log(\varepsilon_{t-1}^C) + \eta_t^C \quad \text{with } \eta_t^C \sim \mathcal{N}(0, \sigma_C^2) \\ \log(\varepsilon_t^I) &= \rho_I \log(\varepsilon_{t-1}^I) + \eta_t^I \quad \text{with } \eta_t^I \sim \mathcal{N}(0, \sigma_I^2) \end{aligned}$$

1.5 Summary of the model

$$\begin{aligned}
\beta R_t &= \mathbb{E}_t \left\{ \frac{\varepsilon_t^C}{\varepsilon_{t+1}^C} \left(\frac{C_{jt+1}}{C_{jt}} \right)^{\sigma_C} \right\} \\
W_t &= \chi H_t^{\sigma_L} C_t^{\sigma_C} \\
\frac{R_t}{\varepsilon_t^I} &= \mathbb{E}_t \left\{ \frac{(1-\delta)}{\varepsilon_{t+1}^I} + Z_{t+1} \right\} \\
K_t &= \varepsilon_t^I I_t + (1-\delta) K_{t-1} \\
Y_t &= \varepsilon_t^A K_{t-1}^\alpha H_t^{1-\alpha} \\
Z_t &= \alpha \frac{Y_t}{K_{t-1}} \\
W_t &= (1-\alpha) \frac{Y_t}{H_t} \\
Y_t &= C_t + I_t + g^y \bar{Y} \varepsilon_t^G \\
\log(\varepsilon_t^A) &= \rho_A \log(\varepsilon_{t-1}^A) + \eta_t^A \\
\log(\varepsilon_t^G) &= \rho_G \log(\varepsilon_{t-1}^G) + \eta_t^G \\
\log(\varepsilon_t^C) &= \rho_C \log(\varepsilon_{t-1}^C) + \eta_t^C \\
\log(\varepsilon_t^I) &= \rho_I \log(\varepsilon_{t-1}^I) + \eta_t^I
\end{aligned}$$

1.6 Calibration

Our calibration is standard with the RBC literature.

Description	Name	Value
Discount factor	β	0.99
Physical capital depreciation rate	δ	0.025
Capital share in production	α	0.36
Consumption Risk	σ^C	1
Labour disutility	σ^L	1
Public spending to gdp ratio	$\bar{G}/\bar{Y}$	0.20
Share of hour worked per day	$\bar{H}$	1/3
Shocks' AR(1) term	$\rho^A, \rho^G, \rho^C, \rho^I$	0.95
Shocks' standard deviation	$\eta_t^A, \eta_t^G, \eta_t^C, \eta_t^I$	1%

1.7 Writing the non-linear model

Endogenous and exogenous variable declarations are left unchanged, the same applies for the steady state computation. Concerning the model's block, equations require to be written in a non-linear fashion:

```

1  beta*r = (e_c/e_c(+1)) * (c(+1)/c)^sigmaC;
2  w = chi*(h^sigmaL)*(c^sigmaC);
3  r/e_i = (1-delta)/e_i(+1) + z(+1);
4  e_i*i = k-(1-delta)*k(-1);
5  y = e_a*(k(-1)^alpha)*(h^(1-alpha));

```

```

6 z=alpha*y/k(-1);
7 w=(1-alpha)*y/h;

```

To be interpretable, it is now common to express our endogenous variables in terms of percentage deviation from the steady state (the same way as we did with log-linearisation methods). Letting X_t denotes a endogenous variable, $\bar{X}$ its steady state and $\hat{x}_t$ the percentage deviation from the steady state, we can simply compute $\hat{x}_t$ via:

$$\hat{x}_t = \log\left(\frac{X_t}{\bar{X}}\right).$$

We modify accordingly the code by adding the linear version of the variables still in the model's block:

```

1 ln_y = log(y/STEADY_STATE(y));
2 ln_c = log(c/STEADY_STATE(c));
3 ln_k = log(k/STEADY_STATE(k));
4 ln_i = log(i/STEADY_STATE(i));
5 ln_h = log(h/STEADY_STATE(h));
6 ln_w = log(w/STEADY_STATE(w));
7 ln_r = log(r/STEADY_STATE(r));
8 ln_z = log(z/STEADY_STATE(z));

```

We use here the function `STEADY_STATE(.)`, it is an internal function in Dynare which returns the steady state value of the variable called in the function.² We are done with the model's block.

To linearise the model, Dynare requires the value (or at least the guess value) of the steady state. Dynare has a built-in solver which works fairly well for relatively small models, nonetheless I advise you to avoid using Dynare's solver to find the steady state as it can find corner solutions which will give fallacious results (*e.g.* a steady state full of zeros).

Use the command `initval;` to provide to Dynare the steady state of the model:

```

1 initval;
2 k = Kss;
3 c = Css;
4 y = Yss;
5 h = Hss;
6 z = Zss;
7 w = Wss;
8 h = Hss;
9 r = Rss;
10 i = Iss;

```

²An alternative solution could be:

```

ln_y = log(y/Yss);
ln_c = log(c/Css);

```

....

And requires that `Yss`, `Css`, (etc) to be declared as deep parameters. Dynare is very flexible and allows you to code in many different ways.

```
11 ln_y = 0;  
12 ln_c = 0;  
13 ln_k = 0;  
14 ln_i = 0;  
15 ln_h = 0;  
16 ln_w = 0;  
17 ln_r = 0;  
18 ln_z = 0;  
19 e_a = 1;  
20 e_g = 1;  
21 e_c = 1;  
22 e_i = 1;  
23 end;
```

The rest of the model can be left unchanged. Nonetheless it is interesting to call the command `resid(1)`; right after closing the `initval`'s block, it will return in Matlab's console the residual of each equation of the model. *i.e.* if you have provided the right steady state, residuals will be of all zeros. It is also possible to add the command `steady`; to use Dynare's solver to find the steady state.

```
1 shocks;  
2 var eta_a; stderr 0.01;  
3 var eta_g; stderr 0.01;  
4 var eta_c; stderr 0.01;  
5 var eta_i; stderr 0.01;  
6 end;  
7 resid(1);  
8 steady;  
9 stoch_simul(irf=60) ln_y ln_c ln_k ln_i ln_h ln_w ln_r ln_z;
```

We finally calibrate the shocks and simulate our economy.

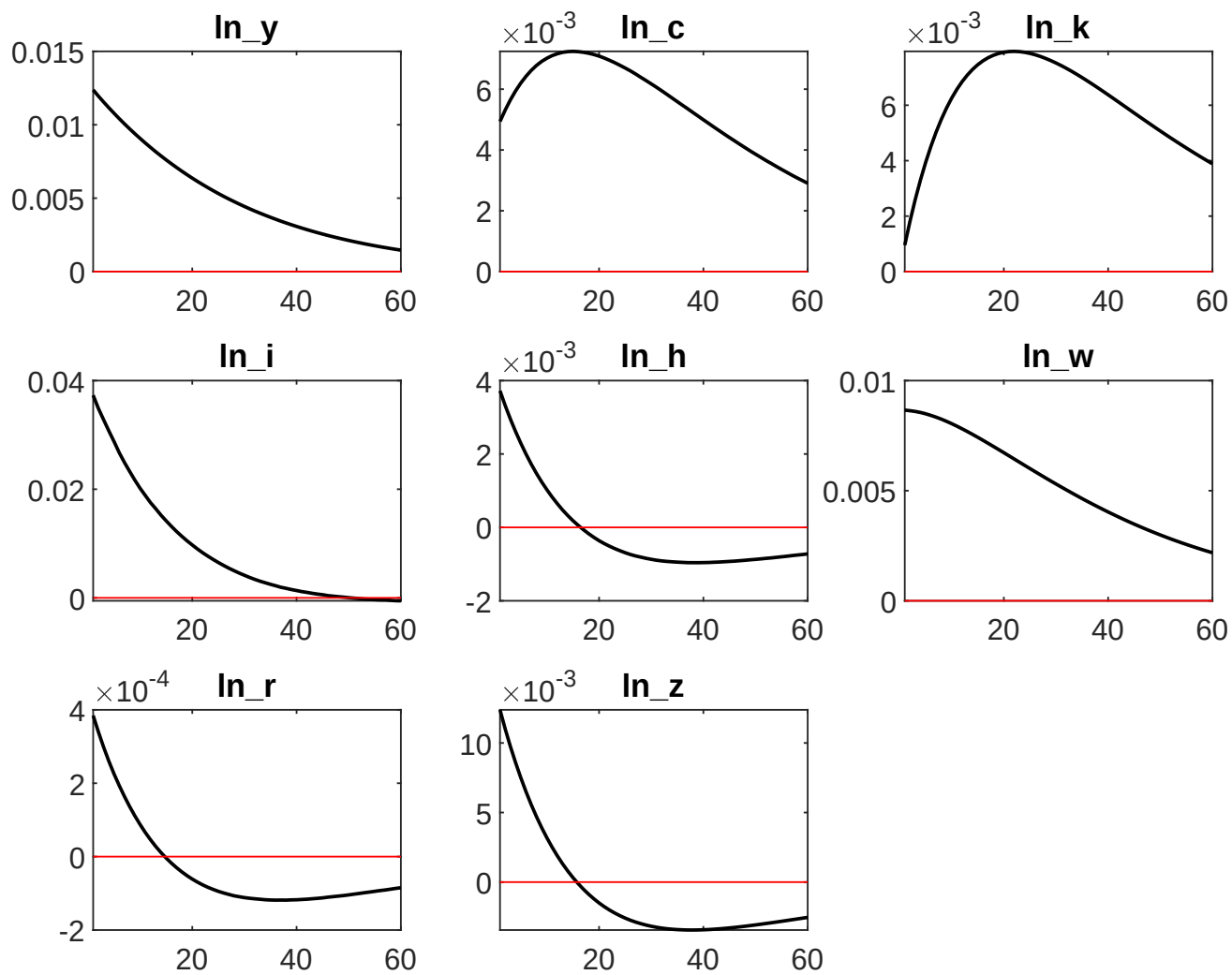


Figure 1: Impulse response function after a 1% productivity shock

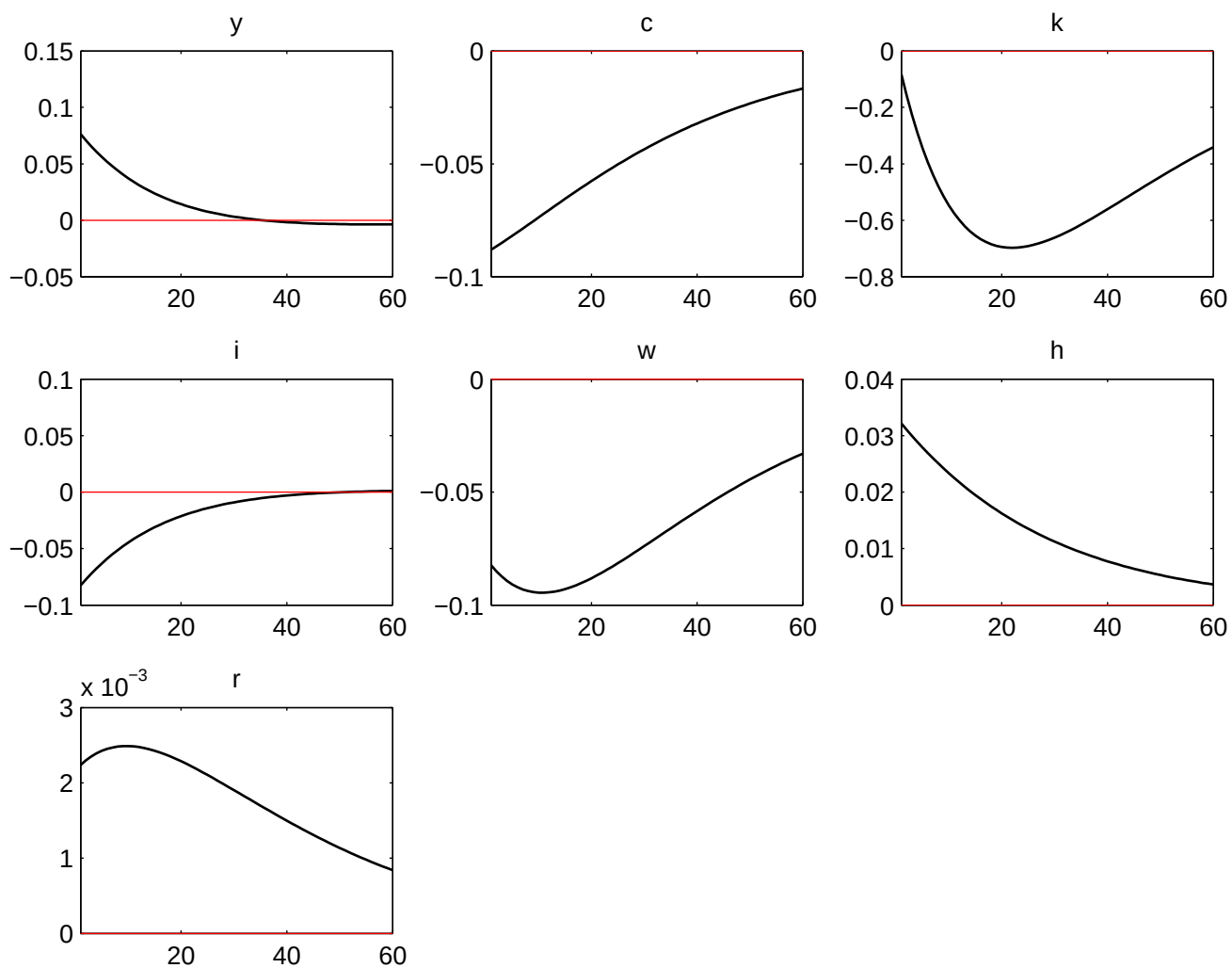


Figure 2: Impulse response function after a 1% spending shock

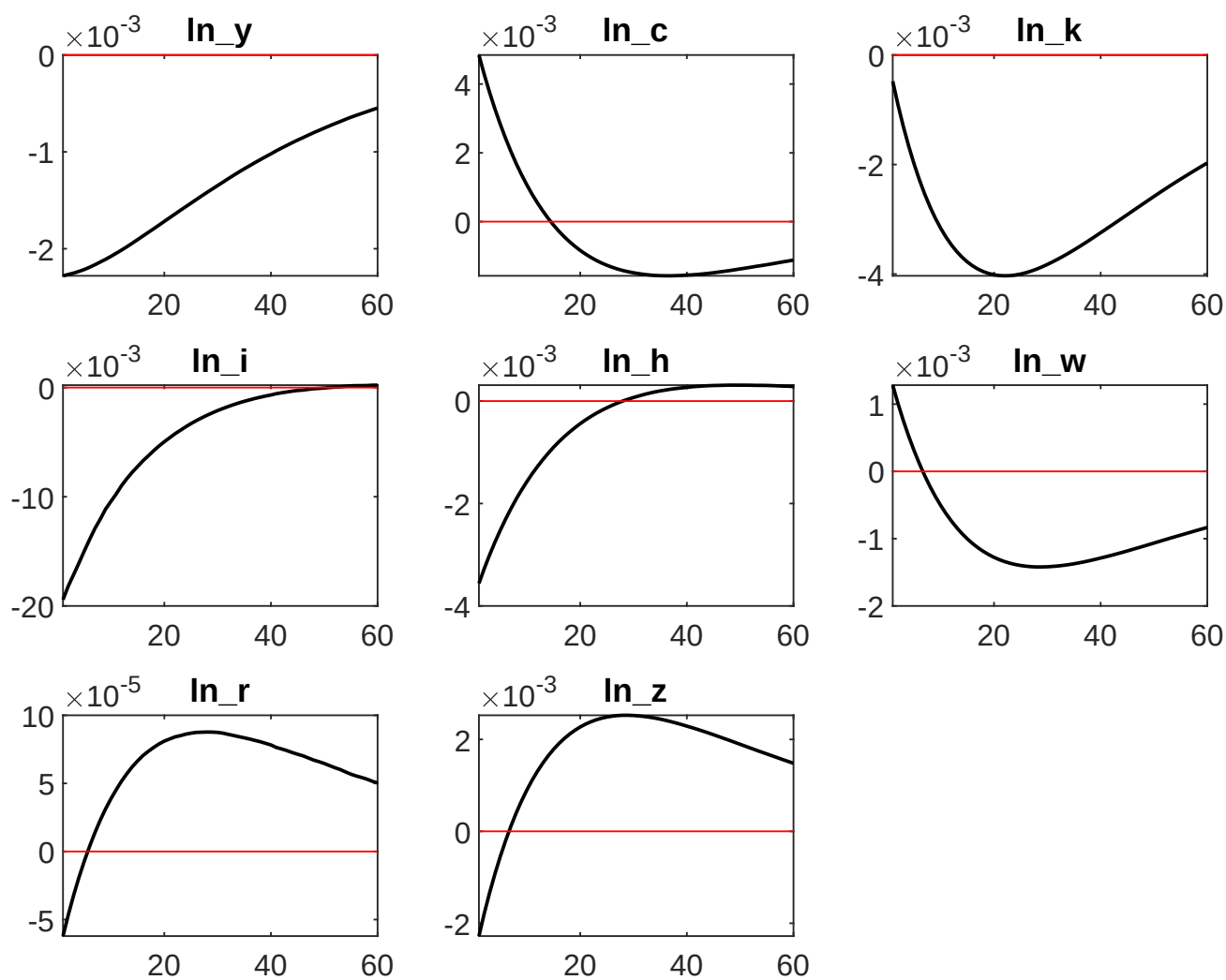


Figure 3: Impulse response function after a 1% preference shock

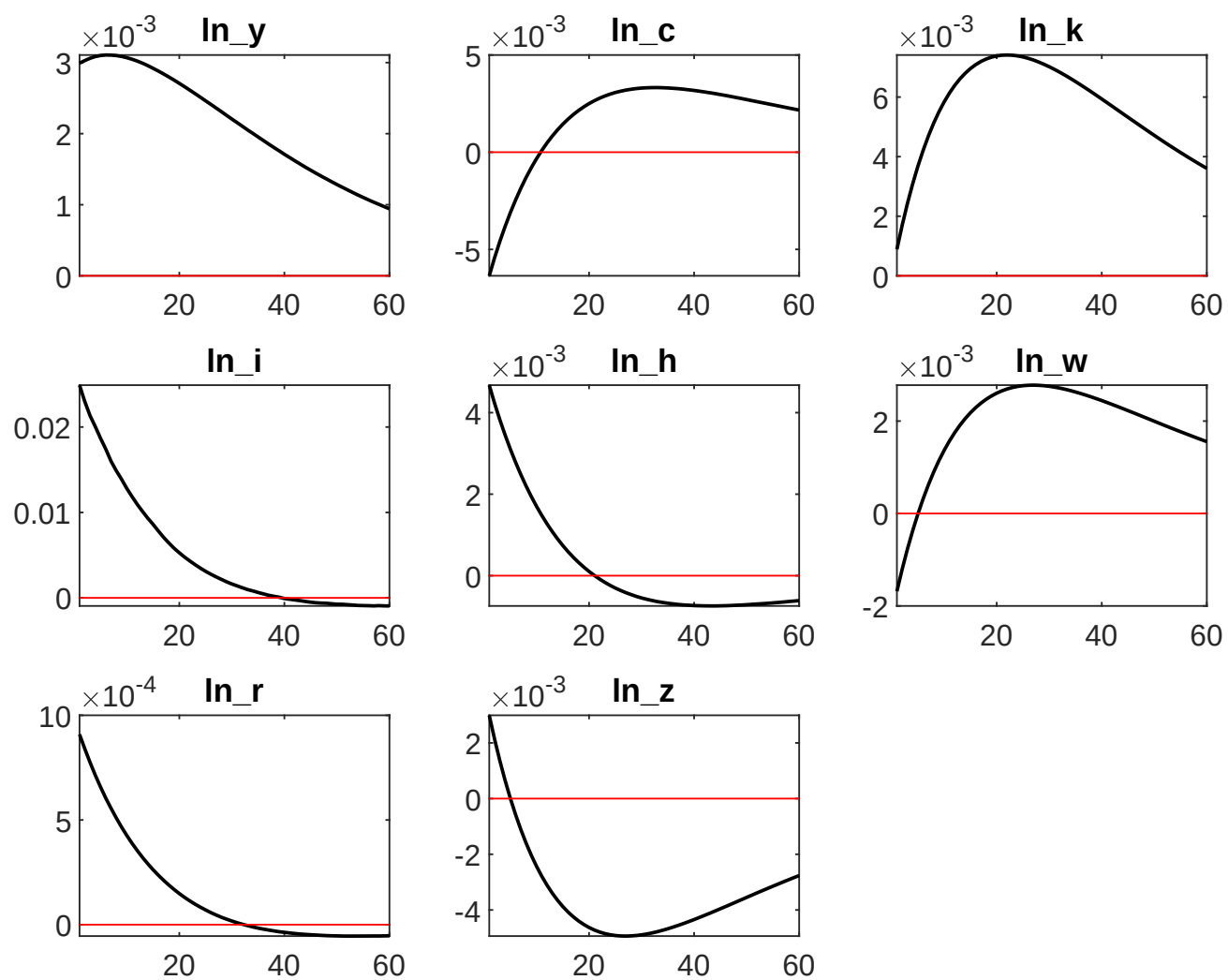


Figure 4: Impulse response function after a 1% investment shock

Exercise 1: Real frictions

Let's consider the baseline RBC model without frictions.

1. Can you show the analytical solution of the steady state? (especially the steady state amount of physical capital).
2. The standard RBC model tends to underestimate the autocorrelation of consumption. To improve the model's fit, it is rather common to assume households face consumption habits. To micro-found such habits, the utility function of households is altered by consumption habits:

$$\mathcal{U}(C_{jt}, C_{t-1}, H_{jt}) = \frac{(C_{jt} - hC_{t-1})^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{jt}^{1+\sigma_L}}{1+\sigma_L}$$

where h denotes the degree of external habits over the consumption of **all** households (this is why subscript j is taken away for C_{t-1}).^a

- (a) Find the modified first order condition.
 - (b) Assuming $h = 0.7$, find the new steady state and plot the system response.
 - (c) Does this real friction significantly increase the autocorrelation of consumption and output?
 - (d) Compare the IRF of the model with and without consumption habits by plotting their IRFs on a single figure.
3. Following [Christiano et al. \(2005\)](#), it is also possible to introduce variable capital utilisation in the production technology of firms:

$$Y_{it} = \varepsilon_t^A (K_{it}^u)^\alpha H_{it}^{1-\alpha}, \quad (11)$$

such that K_{it}^u denotes the capital utilised by the firms:

$$K_{it}^u = u_t K_{it-1}, \quad (12)$$

where u_t denotes the utilised level of capital with $\bar{u} = 1$, K_t is the amount of physical capital supplied by households. The utilisation rate may vary over the cycles: if $u_t > 1$, capital is over-utilised, which will force households to increase the capital supply, while if $u_t < 1$, capital is under-utilised.

Turning to households, they choose the optimal level of capital utilisation, u_{jt} . Over-utilisation of capital is costly for households in a proportion $\Phi(u_t)$:

$$\Phi(u_t) = \bar{Z}(u_t - 1) + \frac{1}{2} \frac{\psi}{1-\psi} \bar{Z}(u_t - 1)^2 \quad (13)$$

where $\bar{Z}$ is the steady state cost of capital services and ψ is the elasticity of capital utilisation. The budget constraint of households is now:

$$W_t H_{jt} + Z_t u_t K_{jt-1} + R_{t-1} B_{jt-1} = C_{jt} + I_{jt} + \Phi(u_t) K_{jt-1} + B_{jt} + T_{jt}, \quad (14)$$

Finally, the resource constraint is also affected by the utilisation cost paid in terms of goods:

$$Y_t = C_t + I_t + \bar{G} \varepsilon_t^G + \Phi(u_t) K_{t-1} \quad (15)$$

- (a) Find the first order condition.
- (b) Does this friction affect the steady state of the model? Why?
- (c) Assuming $\psi = 0.7$, how is this model affected by variable capital utilisation?

^aThese habits are called **external** because households are not optimizing over past consumption. There is a large literature which deals with consumption microfoundations. These habits can be external, internal, additive, etc.

Exercise 2: Alternative asset pricing models

It is possible to derive from the standard RBC model the endogenous asset price denoted $Q_t = \lambda_t^k / \lambda_t^c$, defined as the ratio between the marginal cost of raising new capitals (Lagrangian multiplier on capital accumulation constraint λ_t^k) and the marginal utility of consumption (Lagrangian multiplier on budget constraint λ_t^c). Variable Q_t also refers to the well known Tobin's Q published in [Tobin \(1969\)](#). Tobin's Q is the ratio between a physical asset's market value and its replacement value, this ratio has considerable macroeconomic implications as it constitutes the standard bridge between financial markets and markets for goods and services. In steady state, the ratio must be normalised to one to be consistent with asset pricing theory, such that $\bar{Q} = 1$. In our set-up, the capital accumulation constraint was:

$$K_{jt} = \varepsilon_t^I I_{jt} + (1 - \delta) K_{jt-1}, \quad (16)$$

The first order condition in I_{jt} was:

$$(\partial I_{jt}) : \lambda_t^c = \varepsilon_t^I \lambda_t^k. \quad (17)$$

So the implicit asset price value was totally exogenous:

$$Q_t = \frac{\lambda_t^k}{\lambda_t^c} = \frac{1}{\varepsilon_t^I}. \quad (18)$$

1. Edit your dynare file to include the asset price. What is the main driving shock? Do you find this model consistent with the data?
2. Let us assume first that installing new physical capital is costly for capital suppliers. We introduce the following capital adjustment cost function in the budget constraint of households:

$$AC_{jt}^k = \frac{\chi^K}{2} (K_{jt} - K_{jt-1})^2, \quad (19)$$

such that the budget constraint now looks like:

$$W_t H_{jt} + Z_t K_{jt} + R_{t-1} B_{jt} = C_{jt} + I_{jt} + AC_{jt}^k + B_{jt+1} + T_{jt},$$

And the resource constraint now reads as follows:

$$Y_t = C_t + I_t + \bar{G} \varepsilon_t^G + AC_t^k. \quad (20)$$

- (a) Develop the new model by finding the modified first order condition in physical capital.
 - (b) Draw the system response assuming $\chi^K = 5$.
 - (c) Analyse the driving force and the variance decomposition of asset prices under the capital adjustment cost function.
3. Let's consider an alternative framework as initially developped in [Bernanke et al. \(1999\)](#) which replicates fairly well the dynamic of investment over the business cycles. They decentralize investment decisions to a new agent, it is a representative capital supplier which produces a new amount of capital K_{t+1} at a market price Q_t . This amount of new capital is thus financed by the repurchase of depreciated capital $(1 - \delta) K_{t-1}$, investment I_{jt} and

additional adjustment costs $\Phi(K_{jt-1}, I_{jt})$ detailed below. The profit maximisation problem for this new agent k is:

$$\begin{aligned} \max_{K_{kt}, I_{kt}} & Q_t (K_{kt} - (1 - \delta) K_{kt-1}) - I_{kt} - \Phi(K_{kt-1}, I_{kt}) K_{kt-1} \\ \text{s.t. : } & K_{kt} = \varepsilon_t^I I_{kt} + (1 - \delta) K_{kt-1} \end{aligned}$$

The cost function is thus increasing in investment, defined as:

$$\Phi(K_{kt-1}, I_{kt}) = \frac{\chi^{IK}}{2} \left(\frac{I_{kt}}{K_{kt-1}} - \delta \right)^2 K_{kt-1} \quad (21)$$

When the investment grows faster than physical capital, capital suppliers pay a quadratic adjustment cost which aims at reflecting congestion costs. Inserting the constraint in the objective function of capital suppliers, the final optimisation problem reads as follows:

$$\max_{K_{kt}, I_{kt}} Q_t \varepsilon_t^I I_{kt} - I_{kt} - \frac{\chi^{IK}}{2} \left(\frac{I_{kt}}{K_{kt-1}} - \delta \right)^2 K_{kt-1} \quad (22)$$

While households trade share at a market price Q_t decided by capital suppliers:

$$W_t H_{jt} + Z_t K_{jt} + R_{t-1} B_{jt-1} = C_{jt} + Q_t [K_{jt} - (1 - \delta) K_{jt-1}] + AC_{jt}^k + B_{jt} + T_{jt} \quad (23)$$

- (a) Find the modified first order conditions for capital suppliers (which decide the optimal level of investment) and for households (which decide the optimal stock of physical capital).
 - (b) Draw the system response assuming as [Christensen and Dib \(2008\)](#) that $\chi^{IK} = 0.5$.
 - (c) Analyse the driving force and the variance decomposition of asset prices under the capital adjustment cost function.
4. Let's start from the frictionless model presented at the beginning of the lecture. As [Smets and Wouters \(2007\)](#), we consider adjustment costs concern investment rather than capital. Let's consider households take again both capital and investment decisions. The capital law of motion is now:

$$K_{jt} = \varepsilon_t^I I_{jt} (1 - AC_{jt}^I) + (1 - \delta) K_{jt-1} \quad (24)$$

where AC_{jt}^I is an investment cost function such that:

$$AC_{jt}^I = \frac{\chi^I}{2} \left(\frac{I_{jt}}{I_{jt-1}} - 1 \right)^2 \quad (25)$$

- (a) Find the modified first order conditions for capital suppliers (which decide the optimal level of investment) and for households (which decide the optimal stock of physical capital).
- (b) Draw the system response assuming as [Smets and Wouters \(2007\)](#) that $\chi^I = 4$.
- (c) Analyse the driving force and the variance decomposition of asset prices under the capital adjustment cost function.

Exercise 3: Higher order approximations

It is thus possible to perform higher order approximations (i.e. considering second order variations and third order variations). Dynare is able to perform such approximations to the policy function by adding in the `stoch_simul(...)` function the following argument: `order=2` or `order=3` for second and third order approximations respectively. The simple baseline model does not incorporate important non-linearities while the incorporation of real frictions tends to increase the gap between the first and higher order approximations.

Consider the baseline RBC model, and add variable capital utilization, consumption habits and investment adjustment costs.

1. Perform a first order approximation and second order approximation and compare your IRFs.
2. Generate artificial series of output under a first and a second/third order approximation, and compare the simulated series to show how the approximation order matters. (tips: before `stoch_simul` function, paste the function `set_dynare_seed(0)` to compare your series on a regular basis).

Exercise 4: Estimation of the RBC model

Consider again the baseline model augmented with real frictions. Download the dataset of the real growth of output, investment and consumption as well as the amount of hours worked for the US economy between 1970 to 2015. Write in the model the following equations:

$$y_t^{obs} = \log \left(\frac{Y_t}{Y_{t-1}} \right)$$

$$c_t^{obs} = \log \left(\frac{C_t}{C_{t-1}} \right)$$

$$i_t^{obs} = \log \left(\frac{I_t}{I_{t-1}} \right)$$

$$h_t^{obs} = \log \left(\frac{H_t}{\bar{H}} \right)$$

Let us assume we have no trend in the data (I have demeaned the dataset), so we can assume there is no trend in the model. Measurement equations must be equal to zero in steady state to be consistent with the demeaned data.

1. Estimate the model and the structural parameters of the models (especially the real frictions we have introduced).
2. What are the driving force of output in the sample period up to one, four and one hundred quarters?
3. What is the variance decomposition of output, hours and investment?
4. Is the model able to replicate the correlation between hours worked and output?
5. How trends would have affected the measurement equations?

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